Which Excel function do I need?

Start with your question. Pick the branch, then the sub-question. Every one of Excel's 111 statistical functions, 21 helpers and 27 tools is an answer below. Rule of thumb: p below α (usually 0.05) means the difference is real, but check normality and equal variance first.

What do you need to know?

01 Describe what I have
02 How likely is it?
03 How sure am I?
04 Is the difference real?
05 Do they move together?
06 Can I model Y from X?
07 What comes next?
08 What shape is it in?
09 What settings hit the target?
10 Want the full output table?
11 Old workbook, old name?
12 What can't Excel do?

01 · DESCRIPTIVE STATISTICS 36
Describe what I have

— WHERE IS THE CENTRE?

- **AVERAGE**(number1, [number2], ...) — Default centre of a data set
- **AVERAGEA**(value1, [value2], ...) — Only when logical values must count
- **AVERAGEIF**(range, criteria, [average_range]) — Mean for one shift, grade or line
- **AVERAGEIFS**(average_range, criteria_range1, criteria1, ...) — Mean by heat and orientation together
- **MEDIAN**(number1, [number2], ...) — Skewed data such as delays and cycle times
- **MODE.SNGL**(number1, [number2], ...) — Discrete readings with repeats
- **MODE.MULT**(number1, [number2], ...) — Data with two or more peaks
- **GEOMEAN**(number1, [number2], ...) — Growth rates, ratios, log-normal data
- **HARMEAN**(number1, [number2], ...) — Averaging rates such as units per hour
- **TRIMMEAN**(array, percent) — A robust mean when outliers are present

— HOW SPREAD OUT IS IT?

- **STDEV.S**(number1, [number2], ...) — Almost every quality calculation on sample data
- **STDEV.P**(number1, [number2], ...) — Only when you hold the whole population
- **STDEVA**(value1, [value2], ...) — Rarely; text becomes 0 and distorts results
- **STDEVPA**(value1, [value2], ...) — Rarely; same caution as STDEVA
- **VAR.S**(number1, [number2], ...) — ANOVA, F-tests and pooled variance
- **VAR.P**(number1, [number2], ...) — Complete populations only
- **VARA**(value1, [value2], ...) — Rarely
- **VARPA**(value1, [value2], ...) — Rarely
- **AVEDEV**(number1, [number2], ...) — A spread summary that resists outliers
- **DEVSQ**(number1, [number2], ...) — Building ANOVA sums of squares by hand

— WHAT SHAPE IS IT?

- **SKEW**(number1, [number2], ...) — Quick normality and symmetry screen
- **SKEW.P**(number1, [number2], ...) — Skewness of a complete population
- **KURT**(number1, [number2], ...) — Screening for heavy or light tails

— WHAT ARE THE EXTREMES?

- **MAX**(number1, [number2], ...) — Range and worst-case checks
- **MAXA**(value1, [value2], ...) — Rarely
- **MAXIFS**(max_range, criteria_range1, criteria1, ...) — Worst reading per grade or line
- **MIN**(number1, [number2], ...) — Range and minimum-spec checks
- **MINA**(value1, [value2], ...) — Rarely
- **MINIFS**(min_range, criteria_range1, criteria1, ...) — Lowest toughness per heat
- **LARGE**(array, k) — Top-k values and order statistics
- **SMALL**(array, k) — Bottom-k values and order statistics

— HOW MANY ARE THERE?

- **COUNT**(value1, [value2], ...) — The sample size for every other formula
- **COUNTA**(value1, [value2], ...) — Checking for text entries in numeric columns
- **COUNTBLANK**(range) — Missing-data audit
- **COUNTIF**(range, criteria) — Defects beyond a spec limit
- **COUNTIFS**(criteria_range1, criteria1, ...) — Contingency tables and DPU by category

02 · PROBABILITY DISTRIBUTIONS 41
How likely is it?

— NORMAL (BELL) CURVE?

- **NORM.DIST**(x, mean, standard_dev, cumulative) — Share of output below a limit, ppm out of spec
- **NORM.INV**(probability, mean, standard_dev) — Percentile limits; simulation with RAND()
- **NORM.S.DIST**(z, cumulative) — p-values for z statistics
- **NORM.S.INV**(probability) — Critical z, sample size, sigma level
- **GAUSS**(z) — Quick area from the centre
- **PHI**(x) — Plotting the bell curve

— STUDENT T?

- **T.DIST**(x, deg_freedom, cumulative) — Lower-tailed t tests
- **T.DIST.RT**(x, deg_freedom) — Upper-tailed t tests
- **T.DIST.2T**(x, deg_freedom) — Two-sided p-value for a t statistic
- **T.INV**(probability, deg_freedom) — One-sided critical values; negative for small p
- **T.INV.2T**(probability, deg_freedom) — Confidence intervals and two-sided tests

— CHI-SQUARE OR F?

- **CHISQ.DIST**(x, deg_freedom, cumulative) — Lower tail of a variance test
- **CHISQ.DIST.RT**(x, deg_freedom) — p-value for chi-square tests with any df
- **CHISQ.INV**(probability, deg_freedom) — Upper limit of a variance confidence interval
- **CHISQ.INV.RT**(probability, deg_freedom) — Chi-square table lookups; variance intervals
- **F.DIST**(x, deg_freedom1, deg_freedom2, cumulative) — Lower tail of a variance-ratio test
- **F.DIST.RT**(x, deg_freedom1, deg_freedom2) — ANOVA and regression p-values
- **F.INV**(probability, deg_freedom1, deg_freedom2) — Lower limit of a variance-ratio interval
- **F.INV.RT**(probability, deg_freedom1, deg_freedom2) — F tables for ANOVA

— COUNTING DEFECTS OR DEFECTIVES?

- **BINOM.DIST**(number_s, trials, probability_s, cumulative) — Defectives in a sample, pass or fail data
- **BINOM.DIST.RANGE**(trials, probability_s, number_s, [number_s2]) — Acceptance-sampling ranges in one step
- **BINOM.INV**(trials, probability_s, alpha) — Acceptance numbers and exact critical counts
- **NEGBINOM.DIST**(number_f, number_s, probability_s, cumulative) — Inspections until a set number of finds
- **HYPGEOM.DIST**(sample_s, number_sample, population_s, number_pop, cumulative) — Sampling small lots without replacement
- **POISSON.DIST**(x, mean, cumulative) — Defects per unit, c- and u-chart probabilities
- **PERMUT**(number, number_chosen) — Counting sequences and orders
- **PERMUTATIONA**(number, number_chosen) — Codes and passwords style counting
- **PROB**(x_range, prob_range, [lower_limit], [upper_limit]) — Custom discrete distributions
- **COMBIN**(number, number_chosen) HELPER — Counting samples where order does not matter
- **COMBINA**(number, number_chosen) HELPER — Counting with replacement

— TIME, LIFE OR SKEWED DATA?

- **EXPON.DIST**(x, lambda, cumulative) — Time between failures, constant failure rate
- **WEIBULL.DIST**(x, alpha, beta, cumulative) — Life data and wear-out reliability
- **LOGNORM.DIST**(x, mean, standard_dev, cumulative) — Repair times, particle sizes, skewed data
- **LOGNORM.INV**(probability, mean, standard_dev) — Percentile limits for skewed data
- **GAMMA.DIST**(x, alpha, beta, cumulative) — Waiting times for several events
- **GAMMA.INV**(probability, alpha, beta) — Percentiles of gamma-shaped data
- **GAMMA**(number) — Building other distributions by hand
- **GAMMALN**(x) — Large factorials without overflow
- **GAMMALN.PRECISE**(x) — Same use, newer algorithm
- **BETA.DIST**(x, alpha, beta, cumulative, [A], [B]) — PERT task times, proportions
- **BETA.INV**(probability, alpha, beta, [A], [B]) — Exact binomial confidence limits

03 · STATISTICAL INFERENCE 2
How sure am I?

- **CONFIDENCE.T**(alpha, standard_dev, size) — Confidence interval for a mean, sigma unknown
- **CONFIDENCE.NORM**(alpha, standard_dev, size) — Confidence interval for a mean, sigma known

04 · HYPOTHESIS TESTS 4
Is the difference real?

- **T.TEST**(array1, array2, tails, type) — Comparing two means
- **Z.TEST**(array, x, [sigma]) — Mean against a target when sigma is known
- **F.TEST**(array1, array2) — Checking variances before choosing T.TEST type
- **CHISQ.TEST**(actual_range, expected_range) — Counts by category; contingency tables

05 · CORRELATION 7
Do they move together?

- **CORREL**(array1, array2) — Strength and direction of a straight-line link
- **PEARSON**(array1, array2) — Legacy sheets; identical result
- **RSQ**(known_y's, known_x's) — Share of variation explained by one x
- **COVARIANCE.S**(array1, array2) — Building r by hand on sample data
- **COVARIANCE.P**(array1, array2) — Matches the ToolPak Covariance tool
- **FISHER**(x) — Confidence interval on r; comparing two r values
- **FISHERINV**(y) — Second half of the r interval

06 · REGRESSION 7
Can I model Y from X?

- **LINEST**(known_y's, [known_x's], [const], [stats]) — Simple or multiple regression with inference
- **SLOPE**(known_y's, known_x's) — Change in y per unit of x
- **INTERCEPT**(known_y's, known_x's) — y when x is zero
- **STEYX**(known_y's, known_x's) — Width of prediction intervals
- **TREND**(known_y's, [known_x's], [new_x's], [const]) — Fitted values, residuals, predictions
- **LOGEST**(known_y's, [known_x's], [const], [stats]) — Growth or decay curves
- **GROWTH**(known_y's, [known_x's], [new_x's], [const]) — Predictions from an exponential model

07 · FORECASTING 6
What comes next?

- **FORECAST.LINEAR**(x, known_y's, known_x's) — Straight-line trend forecasts
- **FORECAST**(x, known_y's, known_x's) — Older workbooks only
- **FORECAST.ETS**(target_date, values, timeline, [seasonality], [data_completion], [aggregation]) — Time series with trend and seasonality
- **FORECAST.ETS.CONFINT**(target_date, values, timeline, [confidence_level], [seasonality], [data_completion], [aggregation]) — Error bands around ETS forecasts
- **FORECAST.ETS.SEASONALITY**(values, timeline, [data_completion], [aggregation]) — Confirming the cycle before trusting a forecast
- **FORECAST.ETS.STAT**(values, timeline, statistic_type, [seasonality], [data_completion], [aggregation]) — Judging forecast accuracy

08 · EXPLORATORY DATA ANALYSIS 18
What shape is it in?

— WHERE DOES A VALUE SIT?

- **FREQUENCY**(data_array, bins_array) — Histograms and goodness-of-fit counts
- **QUARTILE.EXC**(array, quart) — Box plots that match Minitab
- **QUARTILE.INC**(array, quart) — Matching older Excel workbooks
- **PERCENTILE.EXC**(array, k) — P5 or P95 limits that match Minitab
- **PERCENTILE.INC**(array, k) — Percentiles that include 0 and 1
- **PERCENTRANK.EXC**(array, x, [significance]) — Where one value sits in the data
- **PERCENTRANK.INC**(array, x, [significance]) — Same, inclusive method
- **RANK.AVG**(number, ref, [order]) — Normal probability plots and rank tests
- **RANK.EQ**(number, ref, [order]) — League tables and Pareto ordering
- **STANDARDIZE**(x, mean, standard_dev) — Flagging points beyond three standard deviations

— HOW DO I SLICE AND SUMMARISE?

- **GROUPBY**(row_fields, values, function, [field_headers], ...) HELPER — Stratify by heat, shift, line or supplier
- **PIVOTBY**(row_fields, col_fields, values, function, ...) HELPER — Side-by-side comparisons such as LPA vs TWA
- **PERCENTOF**(data_subset, data_all) HELPER — Pareto-style contributions
- **FILTER**(array, include, [if_empty]) HELPER — Slicing a subgroup without copying
- **UNIQUE**(array, [by_col], [exactly_once]) HELPER — Listing categories before grouping
- **SORT**(array, [sort_index], [sort_order], [by_col]) HELPER — Ordering data for plots and ranks
- **SUBTOTAL**(function_num, ref1, [ref2], ...) HELPER — Summaries that follow a filter
- **AGGREGATE**(function_num, options, ref1, [ref2], ...) HELPER — Robust summaries on messy data

09 · WHAT-IF & OPTIMISATION 18
What settings hit the target?

— WHICH INPUT HITS MY TARGET?

- **Goal Seek** · Data › What-If Analysis › Goal Seek — Which sigma gives Cpk 1.33?
- **Data Table (one variable)** · Data › What-If Analysis › Data Table (column input cell) — Sensitivity of cost or yield to one input
- **Data Table (two variables)** · Data › What-If Analysis › Data Table (row and column input cells) — Cpk map across mean and sigma
- **Scenario Manager** · Data › What-If Analysis › Scenario Manager — Best, likely and worst cases
- **Solver (Simplex LP)** · Data › Solver › Select a Solving Method: Simplex LP — Blends, product mix, scheduling
- **Solver (GRG Nonlinear)** · Data › Solver › Select a Solving Method: GRG Nonlinear — Response optimization, curve fitting
- **Solver (Evolutionary)** · Data › Solver › Select a Solving Method: Evolutionary — Models with IF, LOOKUP or rounding
- **Forecast Sheet** · Data › Forecast › Forecast Sheet — One-click seasonal forecast

— HOW DO I SIMULATE IT?

- **RAND**() HELPER — Driving simulations through .INV functions
- **RANDARRAY**([rows], [columns], [min], [max], [integer]) HELPER — Thousands of draws in one formula
- **RANDBETWEEN**(bottom, top) HELPER — Random sampling of rows
- **SEQUENCE**(rows, [columns], [start], [step]) HELPER — Input grids for sensitivity tables
- **LET**(name1, name_value1, calculation_or_name2, ...) HELPER — Readable one-cell simulations
- **LAMBDA**([parameter1, ...], calculation) HELPER — Custom test statistics and models
- **MAP**(array1, lambda_or_array) HELPER — Formula-based data tables that recalculate fast

— WHAT BUILDS THE MODEL?

- **SUMPRODUCT**(array1, [array2], ...) HELPER — Objective and constraint cells for Solver
- **MMULT**(array1, array2) HELPER — Regression by matrices; linear models
- **MINVERSE**(array) HELPER — Normal equations for regression by hand

10 · ANALYSIS TOOLPAK 19
Want the full output table?

— COMPARE GROUPS?

- **Anova: Single Factor** · Data › Data Analysis › Anova: Single Factor — Three or more group means
- **Anova: Two-Factor With Replication** · Data › Data Analysis › Anova: Two-Factor With Replication — Two factors with repeats
- **Anova: Two-Factor Without Replication** · Data › Data Analysis › Anova: Two-Factor Without Replication — Randomized block designs
- **t-Test: Paired Two Sample for Means** · Data › Data Analysis › t-Test: Paired Two Sample for Means — Before and after on the same units
- **t-Test: Two-Sample Assuming Equal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Equal Variances — Independent groups, equal spread
- **t-Test: Two-Sample Assuming Unequal Variances** · Data › Data Analysis › t-Test: Two-Sample Assuming Unequal Variances — Independent groups, unequal spread
- **z-Test: Two Sample for Means** · Data › Data Analysis › z-Test: Two Sample for Means — Two means with known variances
- **F-Test Two-Sample for Variances** · Data › Data Analysis › F-Test Two-Sample for Variances — Comparing two variances

— DESCRIBE OR RELATE?

- **Descriptive Statistics** · Data › Data Analysis › Descriptive Statistics — One-click summary
- **Regression** · Data › Data Analysis › Regression — Full regression output
- **Correlation** · Data › Data Analysis › Correlation — Several variables at once
- **Covariance** · Data › Data Analysis › Covariance — Several variables at once
- **Histogram** · Data › Data Analysis › Histogram — Quick frequency table
- **Rank and Percentile** · Data › Data Analysis › Rank and Percentile — Relative standing table

— SIMULATE OR SMOOTH?

- **Random Number Generation** · Data › Data Analysis › Random Number Generation — Static simulated data
- **Sampling** · Data › Data Analysis › Sampling — Drawing an audit sample
- **Moving Average** · Data › Data Analysis › Moving Average — Trend smoothing
- **Exponential Smoothing** · Data › Data Analysis › Exponential Smoothing — Level-only forecasts
- **Fourier Analysis** · Data › Data Analysis › Fourier Analysis — Vibration and periodic signals

11 · LEGACY NAMES 39
Old workbook, old name?

BETADIST → BETA.DIST
BETAINV → BETA.INV
BINOMDIST → BINOM.DIST
CHIDIST → CHISQ.DIST.RT
CHIINV → CHISQ.INV.RT
CHITEST → CHISQ.TEST
CONFIDENCE → CONFIDENCE.NORM
COVAR → COVARIANCE.P
CRITBINOM → BINOM.INV
EXPONDIST → EXPON.DIST
FDIST → F.DIST.RT
FINV → F.INV.RT
FORECAST → FORECAST.LINEAR
FTEST → F.TEST
GAMMADIST → GAMMA.DIST
GAMMAINV → GAMMA.INV
HYPGEOMDIST → HYPGEOM.DIST
LOGINV → LOGNORM.INV
LOGNORMDIST → LOGNORM.DIST
MODE → MODE.SNGL
NEGBINOMDIST → NEGBINOM.DIST
NORMDIST → NORM.DIST
NORMINV → NORM.INV
NORMSDIST → NORM.S.DIST
NORMSINV → NORM.S.INV
PERCENTILE → PERCENTILE.INC
PERCENTRANK → PERCENTRANK.INC
POISSON → POISSON.DIST
QUARTILE → QUARTILE.INC
RANK → RANK.EQ
STDEV → STDEV.S
STDEVP → STDEV.P
TDIST → T.DIST.RT or T.DIST.2T
TINV → T.INV.2T
TTEST → T.TEST
VAR → VAR.S
VARP → VAR.P
WEIBULL → WEIBULL.DIST
ZTEST → Z.TEST

12 · BEYOND EXCEL 12
What can't Excel do?

- **Normality test (Anderson-Darling, Shapiro-Wilk)**
  In Excel: Normal probability plot with NORM.S.INV and RANK.AVG; check SKEW and KURT · Python: stats.anderson, stats.shapiro
- **One- and two-proportion tests**
  In Excel: z statistic with NORM.S.DIST; exact test with BINOM.DIST · Python: proportions_ztest
- **Nonparametric tests (sign, Wilcoxon, Mann-Whitney, Kruskal-Wallis)**
  In Excel: Sign test with BINOM.DIST; rank sums with RANK.AVG · Python: stats.wilcoxon, mannwhitneyu, kruskal
- **Equal variances, three or more groups (Levene, Bartlett)**
  In Excel: One-way ANOVA on absolute deviations from each group median · Python: stats.levene, stats.bartlett
- **Post-hoc comparisons (Tukey)**
  In Excel: Fisher LSD with T.INV.2T; no studentized range in Excel · Python: pairwise_tukeyhsd
- **Unbalanced or multi-factor ANOVA and DOE analysis**
  In Excel: LINEST with coded columns for two-level factorials · Python: statsmodels ols with anova_lm
- **Power and sample size**
  In Excel: Sample-size formula with NORM.S.INV · Python: statsmodels.stats.power
- **Process capability**
  In Excel: Cpk from AVERAGE and STDEV.S, or R-bar over d2 · Python: numpy formulas
- **Control charts**
  In Excel: Limits from AVERAGE and constants, drawn as a line chart · Python: matplotlib chart in the cell
- **Gage R&R**
  In Excel: ANOVA method built from DEVSQ · Python: statsmodels random-effects ANOVA
- **Weibull fitting**
  In Excel: LINEST on the linearized Weibull plot · Python: stats.weibull_min.fit
- **Logistic or Poisson regression**
  In Excel: Solver maximizing the log-likelihood · Python: statsmodels Logit or GLM
- **PY**(python_code, return_type) HELPER — scipy.stats and statsmodels for tests Excel lacks

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